Tanmoy Bhattacharjee

Senior Engineering Manager

Engineering Leadership · AI-native Engineering & Agentic AI · LLM Products · System Architecture & Design · Platform Engineering · People Leadership

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SUMMARY

Senior Engineering Manager with **13+ years** in software development and architecture, including **6+ years managing engineering teams** (people, performance and compensation). Grew from Software Engineer to Senior Engineering Manager at PropertyGuru through **four promotions in under six years**; now lead a **~14-person engineering org across 3 teams** (peak ~21), distributed across Singapore, India and Malaysia, with vendors in Vietnam and Europe. Built 3 teams from scratch and hired 15+ engineers and managers. As a member of the AI advisory board, set AI governance, tooling, budget and guardrails, and deliver AI/LLM products (RAG, MCP, vector databases) to production. Designed multi-tenant platforms serving 6 brands in 4 countries.

SELECTED IMPACT

+7.6% lead volume and **~USD 38K/month saved** from the enquiry revamp · **2–3 deploys/week**, up from one every two weeks · **~30% faster** bug fixes and **15–20% faster** feature delivery with AI tooling · **85% adoption** of AI-driven lead management · **<5%** attrition

CORE COMPETENCIES

Leadership: Engineering Strategy & Vision · Org Design & Scaling · Hiring & Team Building · Coaching & Mentorship · OKRs (6-week cycles) · Budget Ownership · Vendor Management · Build vs Buy · Distributed Teams

AI & Data: LLM Products · Agentic AI · RAG · Vector Databases · MCP · AI Governance & Guardrails · LLM Cost Optimization & Model Routing (Jev) · AI-assisted SDLC · ML · OpenAI · Gemini · Claude · Segment · Braze · Statsig (A/B Testing)

Platform & Delivery: System Design · Multi-tenant Architecture · Microservices · GraphQL · Platform Engineering · Developer Experience (DORA, SPACE) · CI/CD (GitHub Actions, Jenkins) · Kubernetes · Docker · Observability & Incident Response (Datadog, Kibana, Sentry, Squadcast) · SOC 2 & PDPA Compliance

Technology: Java · TypeScript · JavaScript · Node.js · NestJS · React · Next.js · Python · AWS · GCP

PROFESSIONAL EXPERIENCE

Software Engineer → Senior Engineering Manager, APAC | PropertyGuru — Feb 2019 – Present

Senior Engineering Manager, APAC (Sep 2024–Present) · Engineering Manager (Apr 2022–Sep 2024) · Tech Lead managing 2 teams (~10 people), incl. performance and compensation management (Nov 2020–Apr 2022) · Senior Software Engineer (Aug 2020–Nov 2020) · Software Engineer (Feb 2019–Jul 2020). Products: Singapore, Malaysia, Thailand, Vietnam, iProperty Malaysia.

Leadership & Organization

- Lead a **~14-person org across 3 teams** (peak ~21), reporting to the Head of Engineering: two lean teams report directly (Home Owner Solutions, Idea to Market) and one EM-led team (Consumer Conversion & Engagement, 6 members).
- **Built 3 teams from scratch** and **hired 15+ engineers and managers**; on the interview panel, helped design the hiring process and adapt it quickly to AI-related changes.
- Grew talent with **6 internal promotions** to Tech Lead / Lead Engineer and groomed and promoted **1 engineering manager**, keeping attrition **under 5%**.
- Led org design changes (merged teams twice, split one team) across Singapore, India and Malaysia; manage vendors in Vietnam and Europe.
- Set yearly vision and quarterly strategy for the engineering org, executed through **6-week OKR cycles** to move and pivot faster.
- Own budgets for org salaries, AI tooling, mapping solutions and contractors; led vendor and build-vs-buy decisions for the consumer data platform.

AI / LLM Products & AI-native Engineering

- As an **AI advisory board** member, set company AI direction, governance, budget, tooling and guardrails for sensitive data; design, scale and launch AI products quickly to market.
- Drove AI-native development across engineering (custom PropertyGuru MCP, GitHub Copilot, Claude, OpenCode): **~30% faster bug fixes** and **15–20% faster feature delivery**.
- Built production LLM capabilities with **RAG, vector databases, MCP and guardrails**; manage LLM cost with smart model routing using **Jev**, TypeSafe AI's new decision model.
- Launched real-time lead management with AI-driven grouping and alerting: **85% adoption, 93% repeat customer rate (RCR)**.
- Delivered AI listing creation with automated descriptions, image moderation, tagging and enhancement, and auto-generated video with voiceover: **62% adoption, 84% RCR**; **~85% of listing moderation** is now handled by AI.
- Built AI moderation and verification of real estate agents at **72% accuracy**.
- Launched an AI home valuation and agent recommendation system, a big bet planned for monetization; ran AI Search experiments that improved conversion rate per user.

Growth, Conversion & Monetization

- Revamped the enquiry (lead submission) journey, **increasing lead volume by 7.6%** and **saving ~USD 38K per month** by partially moving away from WhatsApp for Business; architected the end-to-end WhatsApp-based lead flow with PropertyGuru in the middle.
- For the IPO, shipped better agent offerings and SEO improvements that outpaced competitors and grew market share, including IPO-critical Listing Performance Insights for agents (**~4% subscription uplift, 1.4x adoption MoM**) and Bahasa Melayu support for Malaysia (**~7% organic traffic growth**).
- Took Home Owner Solutions from **0 to 1**, and scaled teams through the **Agent Monetization hypergrowth** phase as Engineering Manager.
- Improved search and consumer experience: new search (**+2.8% CTR, +1.7% conversion**), relevant search results (**+4% conversion**, WoW), GraphQL consumer profiling (**+2% retention**, WoW), Agent Ratings & Reviews (**+4% consumer acquisition**), Google One Tap sign-in (**+7% logged-in users**, WoW), and report/hide listings for listing quality.
- Increased agent engagement: modernized web and app UI (**+7% agent engagement**, MoM) and real-time listing alerts (**76% adoption, 90% RCR**).
- Run A/B testing with **Statsig** (previously Google Optimize) and targeted testing with Segment and Braze; implemented Segment, GA, FB Pixel, DataXu, Yahoo pixel and Braze for personalized recommendations and life cycle marketing (LCM).

Platform & Engineering Excellence

- Designed and scaled **multi-tenant systems unified across 6 brands in 4 countries**, eliminating custom releases per brand.
- Built a React UI kit on Next.js that raised **deployment frequency from once every two weeks to 2–3 per week**; **~85% of legacy PHP retired**, with the rest planned within two quarters.
- Built central REST microservices for preferences and communication with multiple gateway support.
- Reduced AWS cost by optimizing Kubernetes pod resource requirements; CI/CD on GitHub Actions and Jenkins, observability on Datadog, Kibana and Sentry, on-call and incident response via Squadcast, with SOC 2 and PDPA compliance as hard requirements.

Tech Lead, Full Stack | Philips — Apr 2016 – Jan 2019

- Led a **6-member distributed team**.
- Designed and implemented APIs for patient, care-provider and physician management for Hospital to Home, serving **~80,000 patients** across markets.
- Built a population-health data aggregation platform helping hospitals analyze their value-based care; launched Dec 2018.
- Extended German translation support for the launch of WellCentive in Rostock, Germany, for medical insurance management.

Senior Backend Developer | HomeLane — Feb 2015 – Mar 2016

- Built the pricing engine, a token-based auth service with Google and Facebook OAuth2, and search, list and product-detail APIs with sort, range and size filters.
- Built project service aggregation APIs and APIs generating pre-signed URLs for artifact upload/download to and from AWS.

Backend Developer, Platforms | eBay / PayPal (eBay Inc.) — Jun 2013 – Feb 2015

- Built **batch as a service**, replacing individually built batches per use case with a shared service and reducing idle computation cycles; built a batch service to load raw data into MySQL.
- Managed application-level routing using ESB; archived platform-level functionality as dependency libraries.

PATENT & PUBLICATION

- Patent application (Philips): AI-Enabled Personal Healthcare Companion – an E2E Solution. IP ID 82187; Submission ID 2017ID05305.
- IEEE publication: A Cost Effective Solution for Control of a Mobile Embedded System Using GPRS (ISBN 978-1-4673-1850-1).

EDUCATION

B.E., Computer Science & Engineering, Anna University, 2013 · CGPA 8.95

CERTIFICATIONS

DeepLearning.AI (7 courses) · Google AI (10 badges) · Microservices Software Architecture: Patterns and Techniques · Leadership Skills Training: Become an Inspiring Leader! · Leadership: Practical Leadership Skills

ADDITIONAL

Open Source: swagger-ui-express-updated (npm) · Bug fix: OAuth2-Facebook in SocialAuth plugin · Correction: Virtualenvwrapper (OpenCV prerequisite)

Languages: English, Bengali, Hindi